NOT FOUND

MIL-S-20708/17F
13 March 1989

SUPERSEDING
MIL-S-20708/17E
30 July 1987

MILITARY SPECIFICATION SHEET

SYNCHRO, TORQUE DIFFERENTIAL TRANSMITTER, TYPE 15TDX4C

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Synchros described herein shall consist of this specification and the latest issue of MIL-S-20708.

TABLE I Requirements

Requirement	Value	Unit	Tolerance
Frequency	400	Hz	±1%
Primary Voltage	78 (F)	volts	±1%
Primary Current	215.0	milliamps	maximum
Primary Power	3.5	watts	maximum
Impedance			
Zso	375-525	ohms	min - max
Zrs	113-138	ohms	min - max
Impedance Angle			
Zso	80.0-84.0	degrees	min - max
Zrs	27.0-33.0	degrees	min - max
Transformation Ratio	1.154	------	±2%
Phase Shift (Lead)	5.0	degrees	maximum
Electrical Error	8.0	minutes	maximum
Torque Gradient	0.011	ounce-inches per degree	minimum
Friction Torque	0.05	ounce-inches	maximum
Radial Play	0.0006	inches	maximum
End Play	0.0010	inches	±0.0005
Temperature Rise	20.0	degrees (C)	maximum
Variation of Voltage (+10%) and Freq (-5%)	4.9	watts	maximum

(F) denotes changes

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TABLE II Military part number variant characteristics

Military Part No 1/	A ± 010	B ± 010	G Maximum	Shaft Style	Terminal End	U 2/ + 0000 − 0002
M20708/17-01C	555	500	1 798	Splined	Terminal Board	1872
M20708/17-02C	↓	↓	1.798	Plain & Slotted	Terminal Board	↓
M20708/17-03C	↓	↓	1 536	Plain & Slotted	Wire Leads	↓

1/Military part number M20708/17-01C shall be the NATO standard

1/The suffix letter following the numerical dash number corresponds to the latest modification letter in the type designation

(F) 2/The tolerance for plain and slotted shaft diameters is + 0000 and − 0005

TABLE III Military part number cross-references

SUPERSEDED MILITARY PART NUMBERS		NEW MILITARY PART NUMBERS
MIL-S-20708/17D	MIL-S-20708/17C	
M20708/17D-001	M20708/17-001	M20708/17-01C
M20708/17D-002		M20708/17-02C
M20708/17D-003		M20708/17-03C

NOTE All line items shown in Table III refer to equivalent and interchangeable synchros of the same type designation modification Part number changes do not affect form, fit, or function of the synchros listed therein

Custodians:
Army-AR
Navy-AS
Air Force-85

Review Activities
Army-MI,AV
Navy-OS
Air Force-11,17,99
DLA-ES

User Activities
Navy-MC,CG

Preparing Activity
Navy-AS

(Project 5990-0349-13)

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